

NORAN ESSAM

PMP® | Certified Lean Six Sigma Black Belt

Senior Mechanical Engineer — HVAC Estimation, Operations Leadership & Process Improvement

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PROFESSIONAL SUMMARY

PMP®-certified Mechanical Engineer and Lean Six Sigma Black Belt with 5+ years in HVAC air-distribution manufacturing, spanning technical office, estimation and operations leadership. Led an 18-member cross-functional team handling 9,500+ service requests and 2,000+ work orders a year, built 6 Power BI KPI dashboards, and developed an AI-powered BOQ-to-quotation system and a custom ERP. Combines deep HVAC product knowledge with data-driven process improvement.

CORE COMPETENCIES

Operations & Projects: Project management (PMP®) · Lean Six Sigma (DMAIC, VSM, RCA, Pareto) · KPI design & reporting · Workforce & capacity planning · Risk & escalation management · SOP & workflow design

HVAC Engineering: BOM costing & estimation · Technical submittals & compliance · Product selection (air outlets, dampers, VAVs, louvers, attenuators, ductwork) · Acoustic analysis per AHRI 885

Digital & Tools: Power BI · Advanced Excel · AI automation · ERP design · MS Project · AutoCAD · Revit · SolidWorks · Microsoft 365

PROFESSIONAL EXPERIENCE

Senior Technical Office & Estimation Engineer

Apr 2026 – Present

AL SAMMAN for Air Distribution — Ductwork & Accessories Factory · Egypt

- AI quotation automation:** Built an AI-powered BOQ-to-quotation system that reads BOQs in Excel, PDF, Word or scanned form (Arabic/English) and produces complete quotations, turning a manual process into a semi-automated workflow and significantly cutting turnaround time.
- ERP development:** Designed and developed a customised ERP covering costing, BOMs, quotations, inventory and operational data, improving accuracy and traceability.
- Cost estimation:** Prepare BOM calculations and full pricing for HVAC products: raw material analysis, manufacturing and labour cost estimation.
- Technical coordination:** Own end-to-end technical submittals, prequalification documents and compliance records for complex HVAC projects.

CSD Operations Team Leader

Aug 2024 – Feb 2026

Rabie Al-Takeef Factory — HVAC Manufacturing · Egypt

- Team leadership:** Led and mentored an 18-member cross-functional team of engineers and data specialists managing 9,500+ SSRs, 2,000+ work orders and 500+ quotations a year.
- Data & reporting:** Built and deployed 6 Power BI KPI dashboards (SSR, work orders, quotations, workload, employee performance, reporting), replacing about 8 hours a week of manual Excel reporting.
- Process improvement:** Redesigned SSR and work-order lifecycles using VSM, Pareto and root cause analysis; resolved 20+ critical technical issues and standardised team workflows.
- Risk management:** Acted as primary escalation point for project risks, mitigating 3 major schedule overruns through early intervention and resource reallocation.
- People development:** Led workforce planning, recruiting and onboarding 10 new team members; designed a 7-module HVAC training curriculum and mentored 15+ junior engineers and data specialists.

Technical Support Engineer

Dec 2021 – Jul 2024

Rabie Al-Takeef Factory — HVAC Manufacturing · Egypt

- Prepared technical offers, quotations, submittals and prequalification documents for 200+ proposals across Egypt and the GCC, covering air outlets, dampers, VAVs, louvers, ductwork and sound attenuators.
- Performed full acoustic path analysis per AHRI 885 and selected sound attenuators to meet NC targets on commercial and institutional projects.
- Ensured compliance with SMACNA, ASHRAE 70, AHRI, AMCA 500 and UL 555/555S; ran technical review meetings with consultants and clients.

EDUCATION

M.Sc. Mechanical Engineering — Alexandria University, Egypt

B.Sc. Mechanical Engineering — Alexandria University, Egypt

2021

CERTIFICATIONS

Project Management Professional (PMP)® — PMI · Cert. #4252282

Dec 2025 – Dec 2028

Certified Lean Six Sigma Black Belt — IMC, CPD-accredited (UK), 72 CPD hours · Cert. #C-2026-061167

Sep 2026

Languages: Arabic (native) · English (full professional) · Italian (limited working)